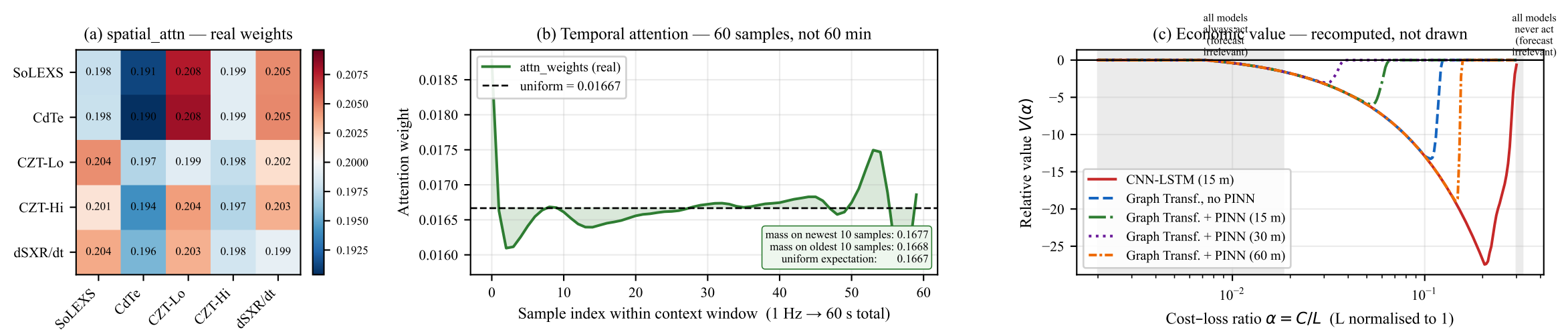
(a) spatial_attn — real weights
SoLEXS
CdTe
CZT-Lo
CZT-Hi
dSXR/dt
0.198 0.191 0.208 0.199 0.205
0.198 0.190 0.208 0.199 0.205
0.204 0.197 0.199 0.198 0.202
0.201 0.194 0.204 0.197 0.203
0.204 0.196 0.203 0.198 0.199
0.2075
0.2050
0.2025
0.2000
0.1975
0.1950
0.1925
range 0.1904–0.2082; uniform = 0.2000
max deviation 4.8 %; rows sum to 1.000, 1.000…
(b) Temporal attention — 60 samples, not 60 min
attn_weights (real)
uniform = 0.01667
Attention weight
0.0185
0.0180
0.0175
0.0170
0.0165
0.0160
mass on newest 10 samples: 0.1677
mass on oldest 10 samples: 0.1668
uniform expectation: 0.1667
0
10
20
30
40
50
60
Sample index within context window (1 Hz → 60 s total)
(c) Economic value — recomputed, not drawn
all models always act (forecast irrelevant)
all models never act (forecast irrelevant)
Relative value $V(\alpha)$
0
−5
−10
−15
−20
−25
CNN-LSTM (15 m)
Graph Transf., no PINN
Graph Transf. + PINN (15 m)
Graph Transf. + PINN (30 m)
Graph Transf. + PINN (60 m)
10^{-2}
10^{-1}
Cost–loss ratio $\alpha = C/L$ (L normalised to 1)